

RUSHI ASHOK SAVANI

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SUMMARY

AI/ML Engineer with experience building machine learning models, LLM applications, and internal decision-support systems in healthcare. Worked across model development, evaluation, retrieval workflows, and cloud deployment, turning complex operational data into tools used by business stakeholders. Strongest in bridging data, AI systems, and production engineering.

EDUCATION

Syracuse University, NY, USA

Masters | Information Systems, Data Science

Aug 2024 - May 2026

(GPA: 3.85/4)

Thadomal Shahani Engineering College, Mumbai, India

Bachelors | Information Technology (*Minors - Artificial Intelligence and Data Science*)

Aug 2020 - Jun 2024

(GPA: 8.52/10)

WORK EXPERIENCE

AI/ML Engineer Co-op

ProHealth Group

Jun 2025 – May 2026

Atlanta GA

- Designed and deployed a healthcare AI assistant that combined **hybrid RAG**, **SQL generation**, and **business-rule validation**, improving response accuracy from **44%** to **92%** for administrator workflows.
- Evaluated **MCP-style** tool orchestration before implementing a custom **LangGraph** workflow to enforce healthcare-specific business rules, SQL validation, and retrieval control across multi-step agent interactions.
- Raised rehospitalization-risk model precision from **64%** to **82%** through leakage audits, temporal feature engineering, SMOTE-Tomek balancing, and **XGBoost** threshold optimization.
- Developed an **LLM-powered anomaly investigation workflow** using Vertex AI, custom SQL pipelines, and grounded clinical-data retrieval, surfacing **~\$70K** in preventable operational and financial losses.
- Productionized ML and LLM services as **Dockerized FastAPI applications on AWS**, adding DeepEval tests, human review, trace logging, caching, and **CI/CD** for reliable multi-user deployment.

Data Scientist - Research Assistant

Syracuse University

Jan 2025 – May 2025

Syracuse, NY

- Fine-tuned a **RoBERTa sentiment classifier** on 5,000+ survey responses with **Hugging Face** and **PyTorch**, reaching 91% accuracy on held-out data through stratified validation and per-class error analysis.
- Processed noisy free-text survey responses with Python **NLP pipelines** and applied **BERTopic**, **Sentence-BERT embeddings**, and **HDBSCAN** clustering to compare AI-adoption sentiment across departments, demographics, and staff levels.

Data Analyst

Krishna Jewels

Jun 2023 - Jun 2024

Mumbai, IN

- Converted paper-based inventory, sales, and supplier records into a structured **MySQL** data layer, migrating 40K+ records and cutting lookup time by **70%** through indexing and query tuning.
- Replaced manual reporting with **SQL ETL pipelines** and **Power BI dashboards** for sales, stock, and margin reviews, saving stakeholders **15+** hours weekly.

TECHNICAL SKILLS

- Programming & Backend:** Python, SQL, TypeScript, FastAPI, REST APIs, Git
- Machine Learning & NLP:** Scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, RoBERTa, Sentence-BERT, Model Evaluation, Feature Engineering
- GenAI & LLM Systems:** LangChain, LangGraph, LlamaIndex, RAG, Prompt Engineering, Vector Search, DeepEval, AWS Bedrock, LiteLLM, Vertex AI, OpenAI
- Cloud & MLOps:** AWS EC2, AWS S3, AWS SageMaker, Docker, GitHub Actions CI/CD, MLflow
- Data Engineering & BI:** PostgreSQL, MySQL, Snowflake, Data Modeling, ETL Pipelines, Power BI, Matplotlib, Streamlit

PROJECTS

PrivateRAG | Private On-premises Q&A System, [GitHub](#)

- Engineered a local RAG backend that ingests private documents, indexes them in Qdrant, and answers questions with Llama 3.1 without relying on external LLM APIs.
- Exposed OpenAI-compatible chat, streaming, and chunk retrieval endpoints with metadata filters, context-window expansion, and source-backed responses for traceable Q&A.

KarmaScore Predictor | Subscription Churn Analysis

- Trained an XGBoost churn model on 30K+ subscriber records, reducing 60 candidate variables to 23 features through recursive feature elimination and SHAP analysis, reaching 72% precision.
- Converted model outputs and SHAP drivers into a retention-risk score, ranking at-risk subscribers in a Streamlit dashboard for intervention planning.

JobFlow AI | LLM-Powered Assistant, [GitHub](#)

- Architected a Gemini-backed FastAPI workflow system that coordinates multi-step LLM generation, input parsing, prompt execution, and document assembly through a reusable backend pipeline.
- Added swappable LLM provider abstraction, concurrent generation, PostgreSQL status tracking, and saved-output management to make generation workflows traceable and easier to scale.